Master's Thesis Research Report

CNN-LSTM Anomaly Detection for Zero-Day & Low-Frequency Intrusions

Comprehensive Dataset Discovery, Schema, Quality & Leakage Audit Report

Date: October 2026 | Author: AI/ML Research Architect | Stage: Read-Only Audit

1. Executive Summary

This report documents the exhaustive empirical audit of two network intrusion benchmark datasets stored within D:\NID Project: CICIDS2017 (Canadian Institute for Cybersecurity) and CSE-CIC-IDS2018 (Communications Security Establishment & CIC). The investigation was conducted in strict read-only mode to establish empirical facts prior to any irreversible preprocessing, feature selection, or CNN-LSTM sequence design.

Key Findings: CICIDS2017 comprises 2,830,743 flows (884.66 MB) across 8 CSV files with a uniform 79-column schema, while CSE-CIC-IDS2018 contains 16,233,002 rows (~6.58 GB) across 10 CSV files with a varying 80-84 column schema. Exactly 78 flow features share a 1-to-1 semantic correspondence between both datasets. Crucially, 10.89% of CICIDS2017 (308,381 flows) are exact duplicate records, and the dataset lacks an absolute Timestamp column. In CSE-CIC-IDS2018, four raw identifier columns ('Flow ID', 'Src IP', 'Src Port', 'Dst IP') were detected in the Thuesday capture.

2. Approved Research Architectural Directives

Based on technical review of the audit findings, the following definitive research decisions have been confirmed:

[DIRECTIVE 1: STRICT DEDUPLICATION MANDATE]

Exact duplicate flows will NOT be tolerated. Duplicate records (308,381 in CICIDS2017; 410,765 in 2018) must be pruned before train/test splitting to eliminate severe data leakage and artificially inflated scores.

[DIRECTIVE 2: DESTINATION PORT EXCLUSION]

'Destination Port' / 'Dst Port' will be removed from the model feature subspace. This prevents shortcut memorization of synthetic service ports (e.g. port 21 for FTP attack) and enforces pure behavioral anomaly learning.

[DIRECTIVE 3: ISOLATED PRIMARY DATASET FOCUS (NO DATASET MIXING)]

CICIDS2017 and CSE-CIC-IDS2018 will NOT be mixed or merged for training. CICIDS2017 remains the primary dataset for model development (aligned with the thesis title), preserving experimental integrity against domain shift.

[DIRECTIVE 4: SEQUENCING & ZERO-DAY DECISION DEFERRAL]

Temporal sequence formulation for CNN-LSTM and specific held-out zero-day attack classes will be finalized in the next stage after establishing the clean, non-leaking feature subspace.

3. Macro Dataset Quantitative Metrics

Metric / Dimension	CICIDS2017	CSE-CIC-IDS2018
Discovered Directory	D:\NID Project\CICIDS2017 Dataset	D:\NID Project\CSE-CIC-IDS2018
Total Files	8 CSV files	10 CSV files
Total Flow Records	2,830,743 flows	16,233,002 (16,232,943 data + 59 repeated headers)
Uncompressed Disk Size	884.66 MB	6,574.61 MB (~6.58 GB)
Column Schema	Uniform 79 columns in all files	Non-uniform (9 files: 80 cols; 1 file: 84 cols)
Shared Flow Features	78 common features (full names)	78 common features (abbreviated names)
Benign Records	2,273,097 (80.30%)	13,484,708 (83.07%)
Attack Records	557,646 (19.70%)	2,748,235 (16.93%)
Exact Duplicate Records	308,381 (10.89%)	410,765 (2.53%)
Timestamp Column	ABSENT (only Flow Duration)	PRESENT ('Timestamp', DD/MM/YYYY HH:MM:SS)
Protocol Column	ABSENT	PRESENT (6=TCP, 17=UDP, 0=HOPOPT)
Raw Network Identifiers	Destination Port only	Dst Port in all; Flow ID, IPs, Src Port in Thuesday
Constant Features	10 features (all 0.0)	10 features (all 0.0)

Testbed Environment	On-premise physical testbed (UNB)	AWS Cloud testbed (Amazon VPC)

4. File-by-File Breakdown & Attack Mapping

A. CICIDS2017 Dataset Files (8 Files)

Filename	Size (MB)	Rows	Duplicates (%)	Primary Attack Classes
Monday-WorkingHours.pcap_ISCX.csv	168.73	529,918	26,935 (5.08%)	100% BENIGN
Tuesday-WorkingHours.pcap_ISCX.csv	128.82	445,909	24,065 (5.40%)	FTP-Patator (7,938), SSH-Patator (5,897)
Wednesday-workingHours.pcap_ISCX.csv	214.74	692,703	81,909 (11.82%)	DoS Hulk (231k), GoldenEye, Heartbleed (11)
Thursday-Morning-WebAttacks.pcap_ISCX.csv	49.61	170,366	6,066 (3.56%)	Brute Force (1,507), XSS (652), SQL Inj (21)
Thursday-Afternoon-Infilteration.pcap_ISCX.csv	79.25	288,602	35,630 (12.35%)	Infiltration (36)
Friday-Morning.pcap_ISCX.csv	55.62	191,033	6,888 (3.61%)	Botnet Ares (1,966)
Friday-Afternoon-PortScan.pcap_ISCX.csv	73.34	286,467	72,353 (25.26%)	PortScan (158,930)
Friday-Afternoon-DDos.pcap_ISCX.csv	73.55	225,745	2,633 (1.17%)	DDoS LOIC (128,027)

B. CSE-CIC-IDS2018 Dataset Files (10 Files)

Filename	Size (MB)	Rows	Cols	Primary Attack Classes
Wednesday-14-02-2018.csv	341.63	1,048,575	80	FTP-BruteForce (193k), SSH-Bruteforce (187k)
Thursday-15-02-2018.csv	358.53	1,048,575	80	DoS GoldenEye (41k), DoS Slowloris (11k)
Friday-16-02-2018.csv	318.26	1,048,575	80	DoS Hulk (461k), DoS SlowHTTPTest (140k)
Thuesday-20-02-2018.csv	3,867.08	7,948,748	84	DDoS attacks-LOIC-HTTP (576k) [Has IP/FlowID]
Wednesday-21-02-2018.csv	313.66	1,048,575	80	DDOS HOIC (686k), LOIC-UDP (1,730)
Thursday-22-02-2018.csv	364.91	1,048,575	80	Brute Force-Web (249), XSS (79), SQL Inj (34)
Friday-23-02-2018.csv	365.11	1,048,575	80	Brute Force-Web (362), XSS (151), SQL Inj (53)
Wednesday-28-02-2018.csv	199.56	613,104	80	Infilteration (68,871)
Thursday-01-03-2018.csv	102.85	331,125	80	Infilteration (93,063)
Friday-02-03-2018.csv	336.04	1,048,575	80	Bot (286,191)

5. Ground Truth Class Distribution & Rarity Audit

A critical requirement of the thesis is evaluating detection of low-frequency and unseen attacks. The class distribution demonstrates severe class imbalance, particularly in CICIDS2017:

Class Label (CICIDS2017)	Count	Percentage	Rarity / Research Designation
BENIGN	2,273,097	80.3004%	Majority Class
DoS Hulk	231,073	8.1630%	Dominant Volumetric Attack
PortScan	158,930	5.6144%	Dominant Scanning Attack
DDoS	128,027	4.5227%	Dominant Flooding Attack
DoS GoldenEye	10,293	0.3636%	Moderate Application Attack
FTP-Patator	7,938	0.2804%	Moderate Brute Force Attack
SSH-Patator	5,897	0.2083%	Moderate Brute Force Attack
DoS slowloris	5,796	0.2048%	Moderate Low-and-Slow Attack
DoS Slowhttptest	5,499	0.1943%	Moderate Low-and-Slow Attack
Bot	1,966	0.0695%	Low-Frequency Attack (Ares Botnet)
Web Attack - Brute Force	1,507	0.0532%	Low-Frequency Web Attack
Web Attack - XSS	652	0.0230%	EXTREMELY RARE ATTACK (<0.05%)
Infiltration	36	0.0013%	EXTREMELY RARE ATTACK (36 samples)
Web Attack - Sql Injection	21	0.0007%	EXTREMELY RARE ATTACK (21 samples)
Heartbleed	11	0.0004%	EXTREMELY RARE ATTACK (11 samples)

6. Data Quality, Artifacts & Leakage Audit

A. Data Quality Anomalies

- Infinity Values: 'Flow Bytes/s' and 'Flow Packets/s' contain +Infinity when Flow Duration is 0 microseconds.
- Missing Values: 'Flow Bytes/s' contains NaNs when both Duration and Byte counts are zero.
- Sentinel -1 Values: 'Init_Win_bytes_forward' and 'Init_Win_bytes_backward' contain millions of -1 values, indicating non-TCP flows or unset initial window negotiation.
- Constant Zero Features: The exact same 10 features have constant value 0.0 across 100% of rows in both datasets (Bwd PSH/URG flags, CWE flag, bulk rate averages). Zero variance.
- Embedded Duplicate Headers: CSE-CIC-IDS2018 contains 59 corrupted rows where column headers are repeated within data rows.

B. Critical Leakage Features Classified

1. Flow ID (Cat A - Critical): Direct 5-tuple identifier. Must be completely excluded.
2. Source & Destination IPs (Cat A/D - Critical): Attacker/victim IP addresses in testbed. Must be completely excluded.
3. Source Port (Cat A - High): Script ephemeral source ports. Must be excluded.
4. Destination Port (Cat E - Approved for Exclusion): Service ports (80, 21, 22) allow models to memorize port numbers rather than network behavior. Exclusion confirmed by researcher.
5. Timestamp (Cat B/D - High): Encodes synthetic daily attack schedules. Must never be used as a numerical feature.

7. Unseen Attack Cross-Dataset Contamination Analysis

If an attack category is held out as unseen/zero-day in CICIDS2017, it must not enter training via another source:
- Heartbleed (11 flows in 2017): Completely absent from CSE-CIC-IDS2018. Pristine unseen candidate.
- PortScan (158k flows in 2017): Absent as distinct label in CSE-CIC-IDS2018. Pristine unseen candidate.
- Infiltration (36 flows in 2017 vs 161,934 flows in 2018): HIGH CONTAMINATION RISK. If Infiltration is held out as unseen on CICIDS2017, CSE-CIC-IDS2018 must NOT be in training.
- DDOS-HOIC and LOIC-UDP: Present only in CSE-CIC-IDS2018; absent from CICIDS2017.

8. Computing Environment & Specifications

- Host Platform: Microsoft Windows 11 Pro 64-bit (Build 10.0.22621)
- Python: 3.11.0 (64-bit AMD64)
- Key Scientific Libraries: PyTorch 2.2.2, Pandas 3.0.0, NumPy 1.26.4, Scikit-Learn 1.8.0, SciPy 1.17.0, FPDF2 2.8.7
- Hardware: Intel(R) Core(TM) i7-7500U CPU @ 2.70GHz (2 physical cores, 4 threads)
- Physical Memory: 8.0 GB Total (~2.0 GB currently available)
- GPU Acceleration: CUDA unavailable (PyTorch running in CPU execution mode)
- Primary Storage: D: Drive with 357.9 GB free capacity